

Trending Topics (trending)

Luca has been hired as an intern at Twitter to improve tools for understanding virality on social media. The company is known for having popularized the now widespread *hashtags*, constituted by the hash symbol # followed by a keyword, used to tag contents (especially tweets, in the case of Twitter).

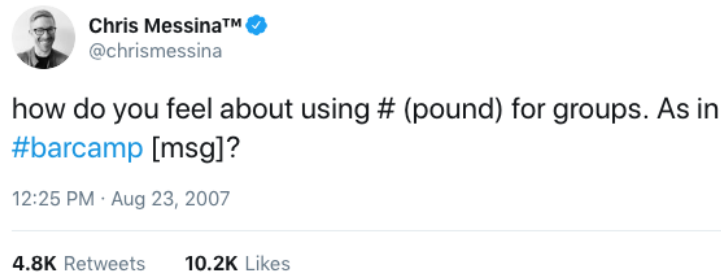


Figure 1: The tweet that made history proposing the use of hashtags in 2007!

Each second, internal Twitter functions report the most used hashtag worldwide in that second (the most used hashtags signal the so-called *trending topics*).

However, this level of granularity can be difficult to comprehend: for this reason, Luca wants to create a tool that aggregates data and reports the most popular hashtag for each given period of time lasting T seconds. For example, when $T = 60$ seconds, the tool should output the most used hashtag in the first minute (seconds 0 to 59, inclusive), then the most used one in the minute that lasts from second 1 to second 60 (inclusive), and so on.

Given the list of the most popular hashtags in each of the last N seconds, are you able to produce such a report?

Among the attachments of this task you may find a template file `trending.*` with a sample incomplete implementation.

Input

The first line contains two integers: N and T . The i -th of the following N lines contains the most popular hashtag in the i -th second.

Output

You need to write a line containing the most popular hashtag *for each period of time that lasts T seconds* (i.e., the first line will contain the most popular hashtag from second 0 to $T - 1$, the second line the most popular hashtag from second 1 to T , and so on).

Constraints

- $1 \leq T \leq N \leq 50\,000$.
- Each hashtag consists of up to 20 lowercase letters of the English alphabet.
- In case of ties, the lexicographically smallest hashtag is chosen.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (6 points) $N \leq 1000, T = 1$.

- **Subtask 3** (24 points) $N \leq 1000, T \leq 10$.

- **Subtask 4** (29 points) $N \leq 5000$.

- **Subtask 5** (27 points) $N \leq 20\,000$.

- **Subtask 6** (14 points) No additional limitations.


Examples

input	output
5 3 ois final ranking ois ois	final final ois
6 4 potus biden trump trump biden capitolhill	trump biden trump

Explanation

In the **first sample case**, we have to consider periods of time of $T = 3$ seconds. In the first of such periods, three different hashtags are observed and therefore the most popular is **final** (due to lexicographical order). In the second period of time, hashtags **final**, **ranking** and **ois** are considered: it is again a tie and **final** is chosen. In the third period of time, **ois** wins because it occurs twice.

In the **second sample case**, we have to consider periods of time of $T = 4$ seconds. In the first of such periods, **trump** occurs twice and is the most popular hashtag. In the second one, **biden** and **trump** both occur twice, but **biden** is preferred due to lexicographical order. In the third one, **trump** occurs twice and is again the most popular hashtag.